Community-driven standards development for reference genome generation

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EUROPEAN REFERENCE GENOME ATLAS

Galaxy EUROPE

How can the Biodiversity Community foster high standards for reference genomes?

ERGA, the European node of the Earth BioGenome Project (EBP) is a pan-continental network committed to cataloguing eukaryotic biodiversity through the generation of high-quality reference genomes.
As a community, we wish to establish and enforce standards on Genome Assemblies and Annotations through robust testing of tools and pipelines. Here, we present methods to evaluate assemblies and the work towards finding gold-standard annotations methods across the tree of life.

Genome assembly review - The ERGA Assembly Report (EAR)

A standardised document, developed with the genomics community, containing all relevant information required to determine the quality of a genome assembly

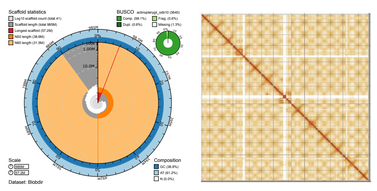
Collate information from species and sequencing data & generate EAR

Upon acceptance, upload genome to ENA

PacBio

ARIMA GENOMICS

Create pull request



Review

https://github.com/ERGA-consortium/EARs/

ENA European Nucleotide Archive

Github Bot assigns reviewer and supervisor from the pool of reviewers in the ERGA community

Reviewer

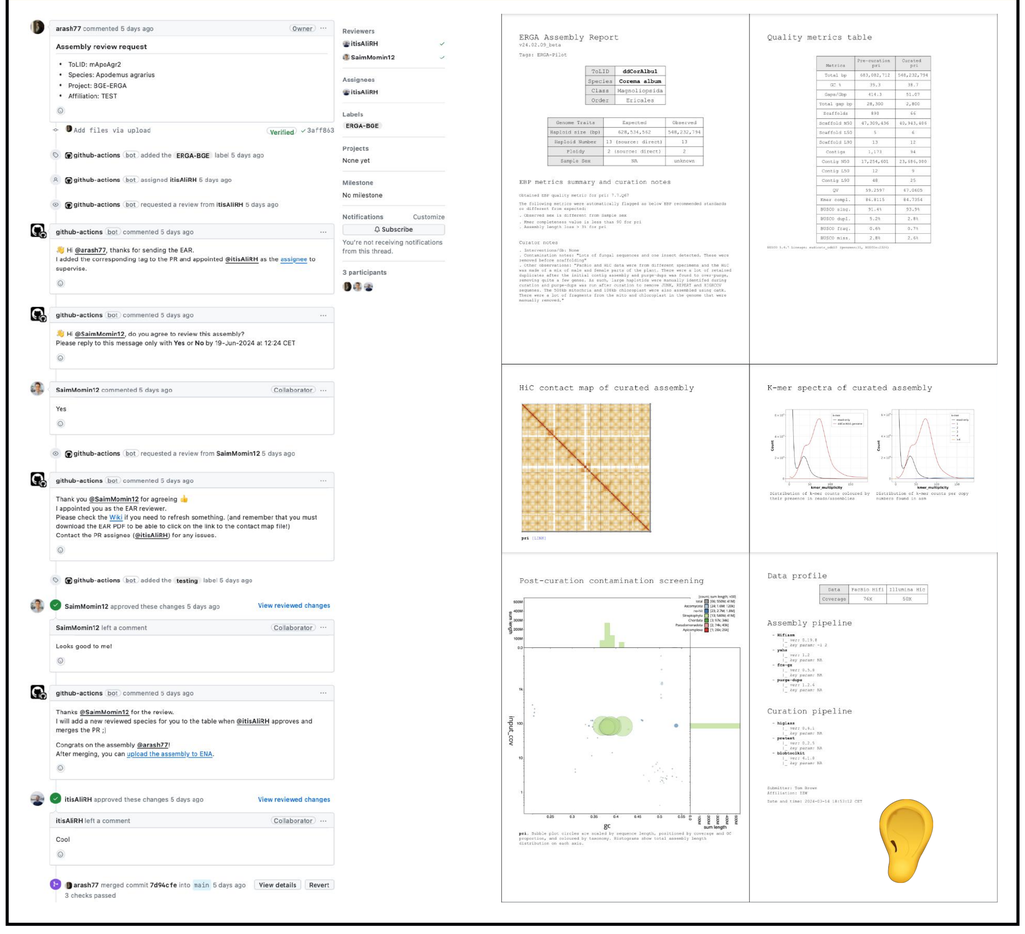
Supervisor

Review genome and request changes

Oversee review process and merge pull request

See the genomes currently reviewed and published by the ERGA community:

Fully documented FAIR review process on GitHub



Create an EAR using Galaxy

ERGA Assembly Reporting Tool (EAR) A tool to compile assembly reports and stastics from assembly pipeline

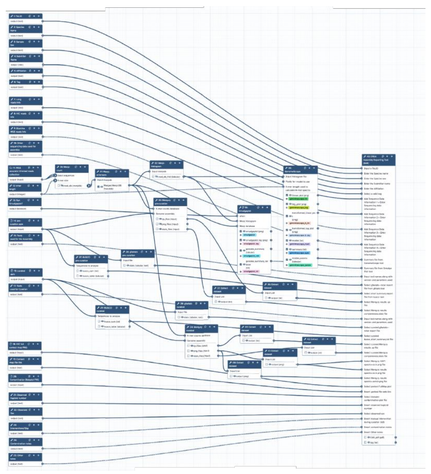
PretextMap to view Hi-C contacts

Blobtools for decontamination

merqury for genome accuracy and completeness

BUSCO Gene Completeness

gfastats for assembly metrics



Galaxy for Genome Annotation

GGA

The expanding repertoire of Galaxy Genome Annotation tools

Repeat Masking

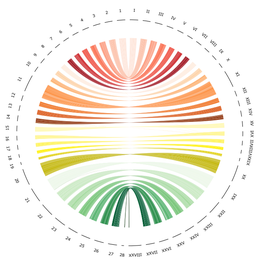
Red

RepeatModeler & RepeatMasker

Visualisation

CIRCOS

JBrowse

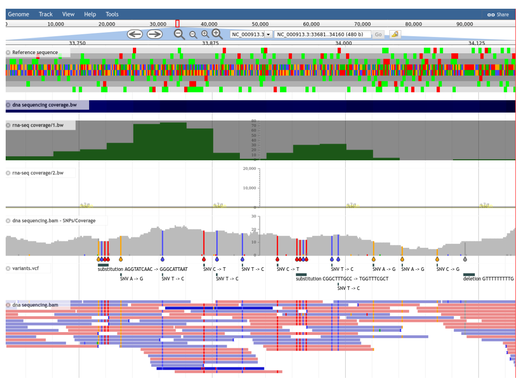


Gene Prediction

MAKER & BRAKER

Helixer

Funannotate



Functional Annotation

InterProScan

Antismash

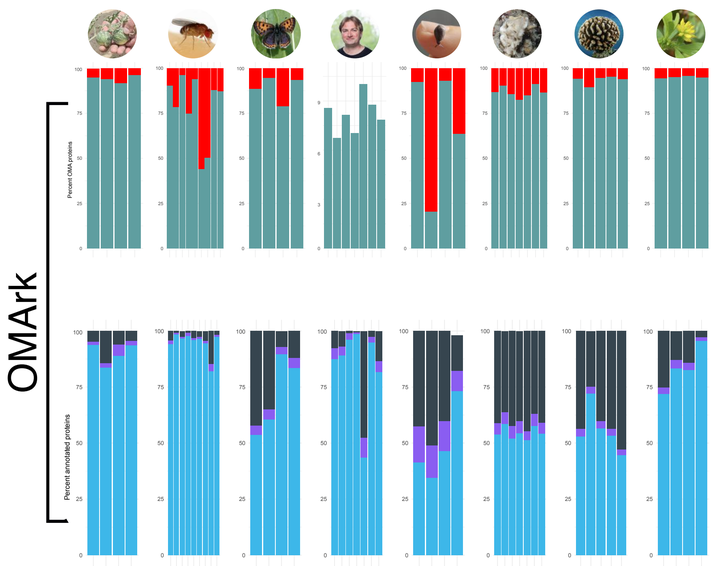
EggNOG-Mapper

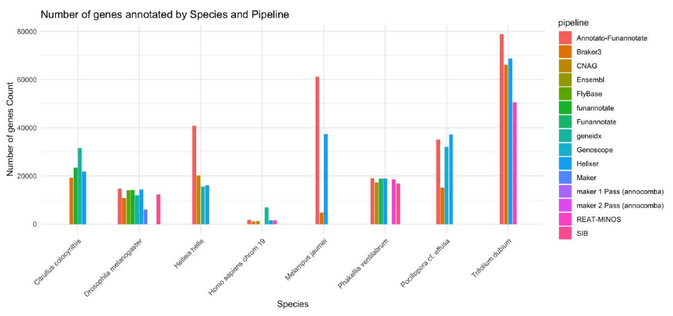
BioHackathon 2023 - Genome Annotation Benchmarking

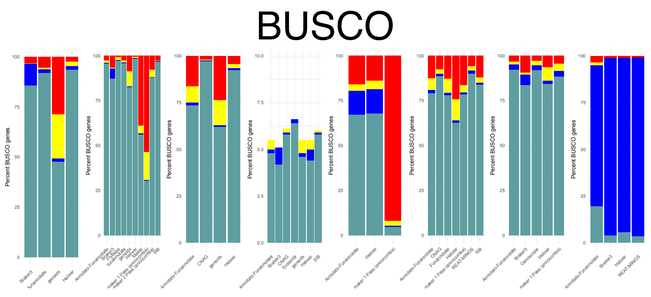
In November 2023, a group of 25 researchers came together at the BioHackathon Europe to benchmark annotation pipelines across a diverse range of species

We created **44** annotations across **8** species using a mixture of in-house pipelines, established annotation tools and cutting-edge software including deep-learning gene prediction

As the number of tools installed in Galaxy and species tested increases, the easier it will become for anyone to annotate any species, regardless of taxon, data availability and access to computational resources







Join our mission!

Become an ERGA member and join our open community meetings:

ERGA Registration
https://www.erga-biodiversity.eu/register

ERGA Pipelines
WorkflowHub

Annotation @ Galaxy
https://annotation.usegalaxy.eu/

BH2023EU BioHackrXiv report
https://doi.org/10.37044/osf.io/fy49g

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